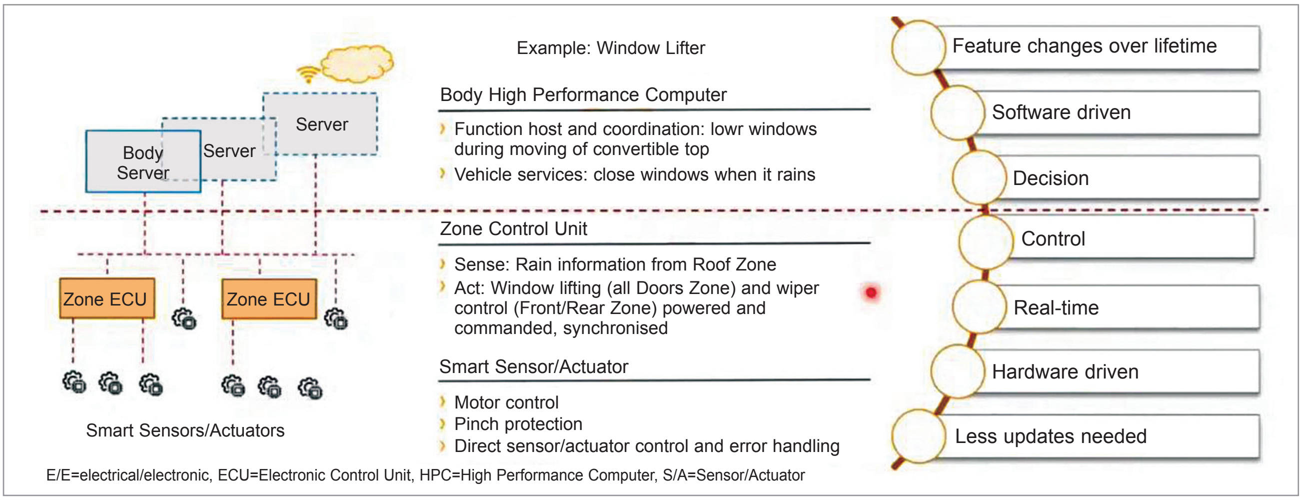

Fig. 5: Example of a window lifter (see text)

ECUs will become bigger and faster. As mentioned, we will see central ECUs and zonal ECUs in architecture.

The question is whether to update all the ECUs or update just the central ECU? Updating each ECU is costly. To avoid this complexity and cost, we have to migrate to a revolutionary approach where only the central ECU is updated and the zonal ECUs stay put with current software.

Next step is to secure the integrated ecosystem. And securing systems within the vehicle only is not sufficient; securing the link between the vehicle and the cloud server is also important. Because advancements in software also lead to high vectors for others to exploit.

Within the vehicle, the ECUs need to be protected from tampering. There must be a system to detect vehicle anomalies and to alert when any kind of attack is happening. The architecture has to be built from ground up considering all these factors.

An example of all these integrations can be seen in Volkswagen ID.3 series (some vehicles have been launched and some are in the pipeline). Basically, in the past, more emphasis was put on the control units, the hardware, the services, and the functions provided. But now, with additional demands, security features have to be built-in as part of vehicle architecture.

Moreover, there must be options for the architecture to talk to the latest IoT devices. These additional demands cause the need for plugins to be already available in the central ECU, zonal ECUs, and other smaller control units.

Regarding connectivity with the cloud, the sensor data has to go through several layers before reaching the cloud and get processed. The sensor senses the data, the central computing device processes the sensed data and transmits it to the cloud. It can be clearly seen that the integration is vertical, therefore we call it vertical integration.

There is horizontal integration where there are multiple domains and third-party software. All of the components in horizontal integration must have plug and play type of product design, so that a particular function can be changed without affecting other components. This kind of integration becomes necessary going forward.

With integrations becoming a bigger aspect, we are talking about a complex system. In order to manage the complex system, there will be an increase in the number of people needed to execute the project. For a typical OEM in Tier 1 cities, this number will rise from 30 to 400. The number of parallel project deployments in an OEM will go up, and as a result the number of departments due to horizontal integration will go up, and so there will be multifold increase in stakeholders.

Further down the line, there will be a twenty-fold increase in the lines of code. Besides, the business model will change. What happens typically is that hardware gets delivered along with some software, but a portion of it or all of the application software may come from third-party vendors. Therefore, we are talking about less share of R&D, which is a clear departure from the past.

What's important is to understand different business units, whether these are internal or external. We have to work together having synergies and establish something like central engineering, which could talk to different stakeholders and bring everything together.

Conclusion

To summarise, here is an actual example of the Volkswagen project that has been recently launched. We observed that there were 68 ECU links in the car. There were 20 protocols to talk to multiple stakeholders, and there were 19 companies aligned for this project. The stakeholders' requirements jumped to around 70,000 and more. And then, in terms of volume of contribution, there were about 600 continental developers.

In order to bring together such an environment, project management has to play a key role. With architecture change, we are paving the road for better connected mobility and software defined vehicles. EFY

The article is an extract from a speech delivered by Vinayaka Nagaraja, Head, Central Engineering - Systems Domain & Body Network Control, Vehicle Networking and Information (VNI), Continental Automotive India, at MOVES 2021. It has been prepared by Darshil Patel, a technology journalist at EFY